



# Overview of the Last Planner® System

Collaborative Construction Progress Planning – Spec. 013200

David Umstot, PE

31 October 2018



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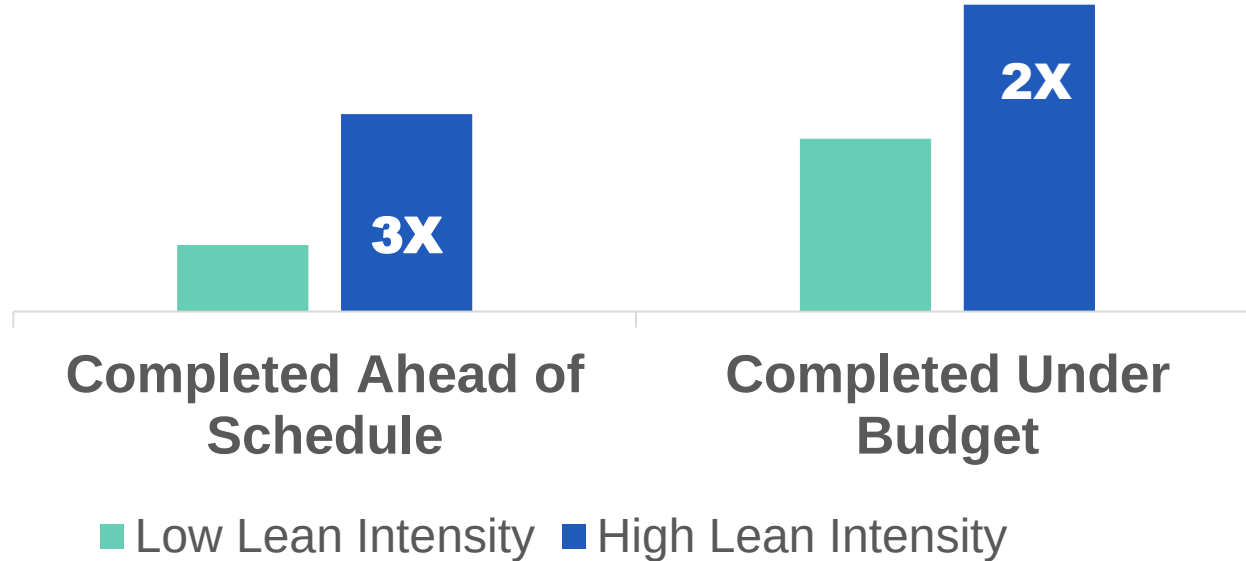
# Sink or Swim Together?

Sure glad the hole isn't at our end.



# Why use Last Planner®?

**DODGE** DATA & ANALYTICS

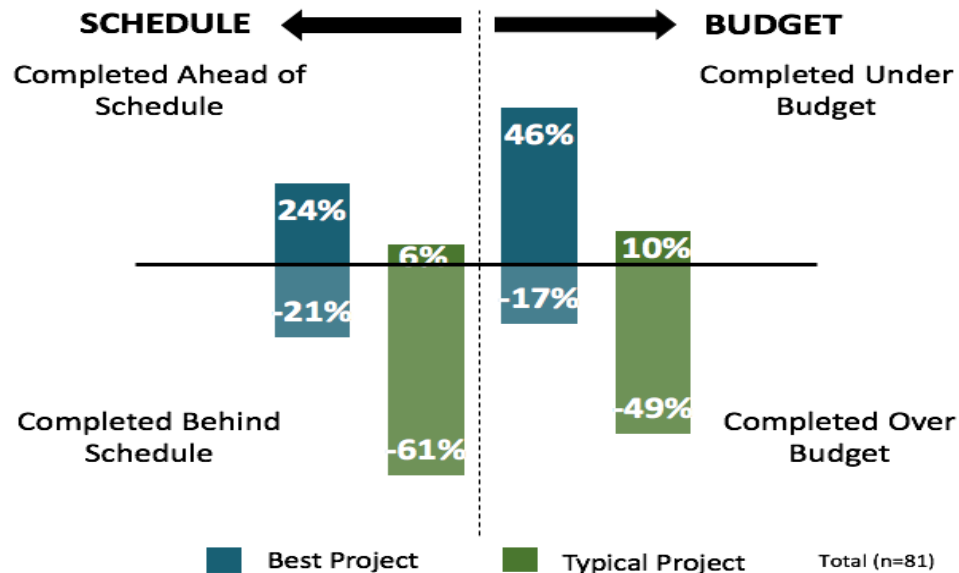


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## Performance from Approval of Capital Project (% of Best/ Typical Projects)



**DODGE** DATA & ANALYTICS



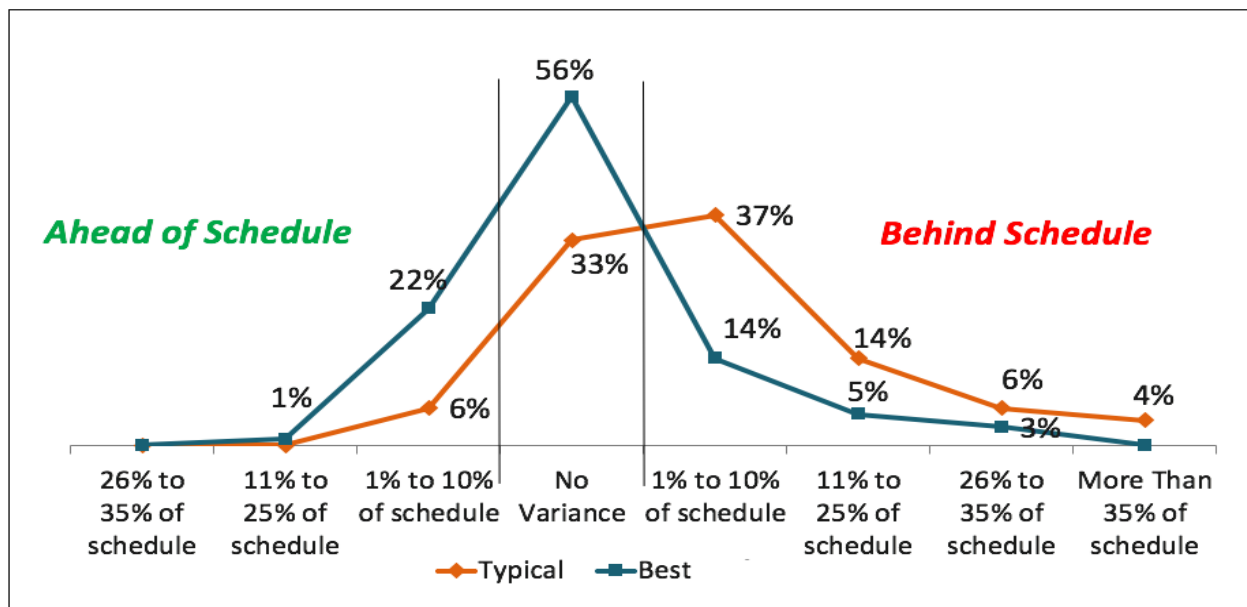
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# Schedule Performance

## Variance of Final Schedule vs. Allocated Capital Schedule



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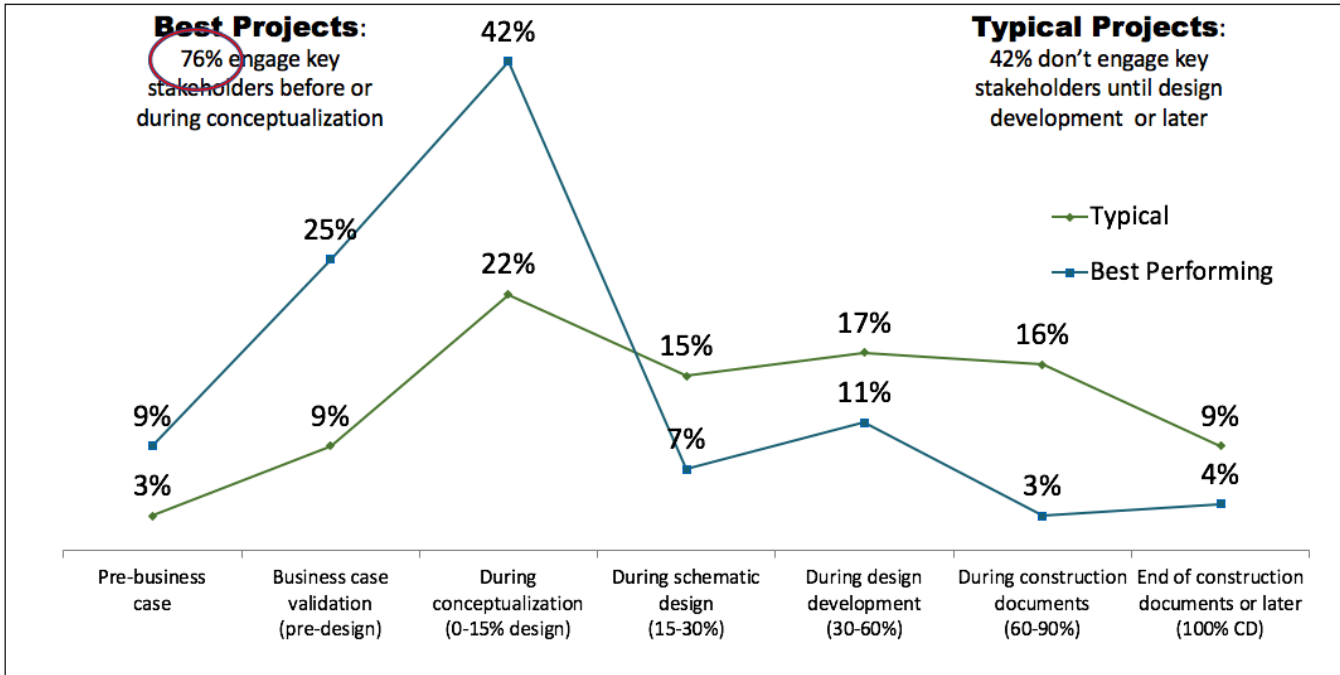


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# Timing of Key Stakeholder Engagement



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# Workflow & Risk

- Workflow losses are real, lead to adversarial relations, and are difficult to quantify, so...
- Everyone protects themselves by adding contingency and/or holding back labor to keep utilization high.
- This further reduces workflow predictability and increases project risk.
- By their/our actions, we increase that risk and shift it along.



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# KEY CONCEPTS:

1. Traditional planning systems are unable to produce predictable workflow.
2. Workflow reliability directly affects system speed and cost.
3. All plans are forecasts, all forecasts are wrong, the longer the forecast the more wrong it is, the more detailed the more wrong it is.



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# Last Planner® System Benefits:

- Improves communication & reliability
- Fosters an enjoyable environment, trust & collaboration
- Promotes early stakeholder engagement
- Improves visibility of the project plan (transparency)
- Creates team buy in
- Rapid learning through metrics, revealing areas for improvement
- Improves planning in both design & construction phases



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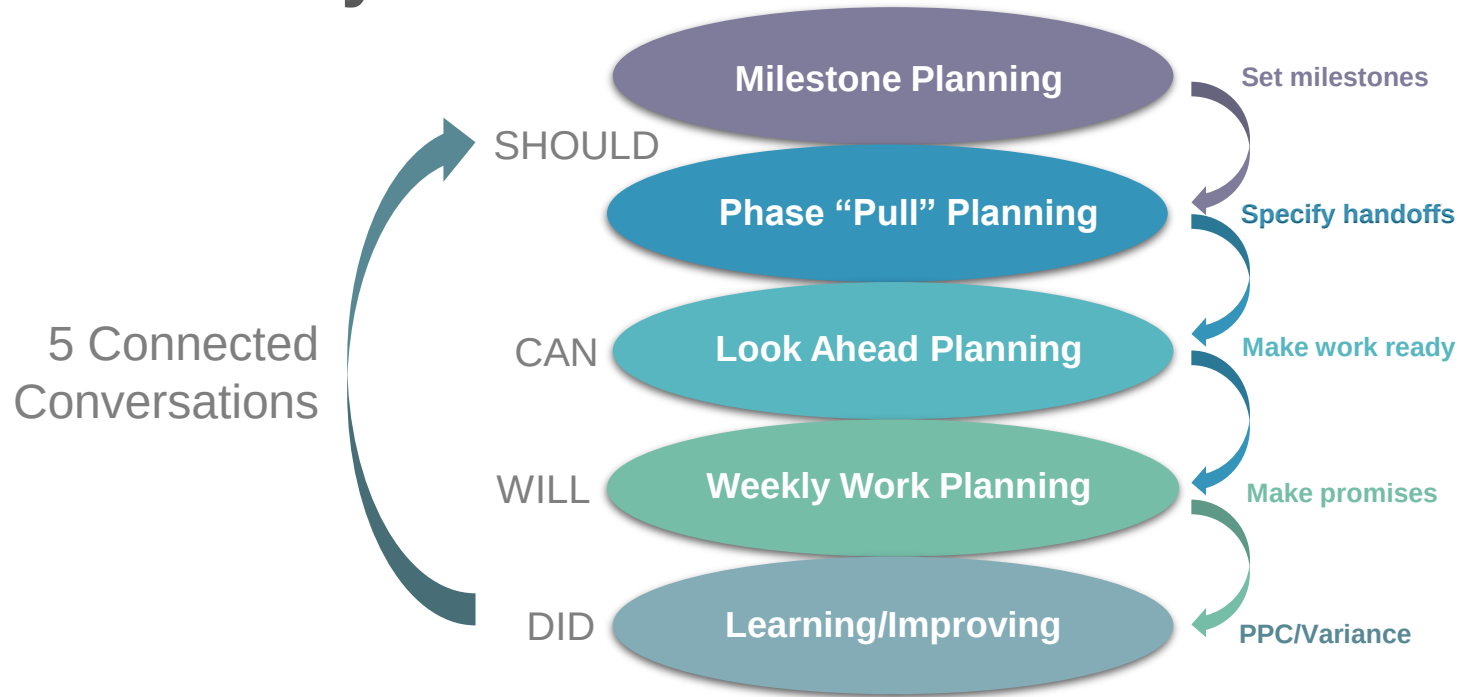
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# Last Planner® System Overview

Hand-offs Matter

# Last Planner<sup>®</sup> System



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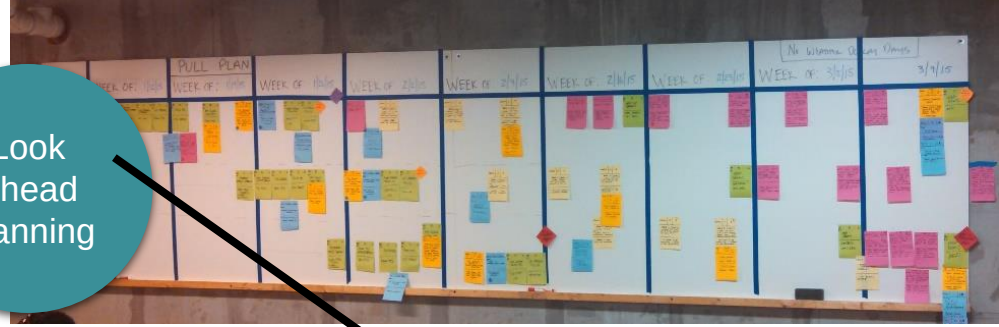
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# Last Planner<sup>®</sup> System

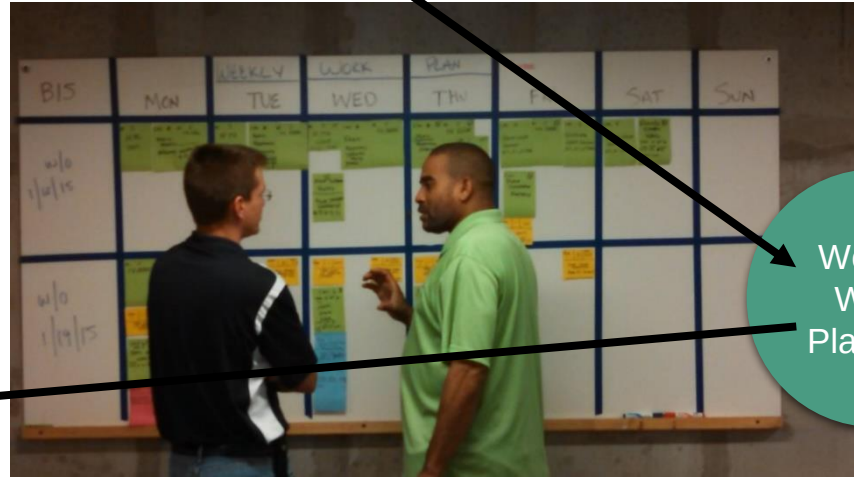


Phase  
Pull  
Planning

Look  
Ahead  
Planning



Daily  
Huddle



Weekly  
Work  
Planning

# Who is the Last Planner?

- Person closest to work, with authority to make decisions creates schedule
- A Last Planner can make the reliable commitment to complete the work

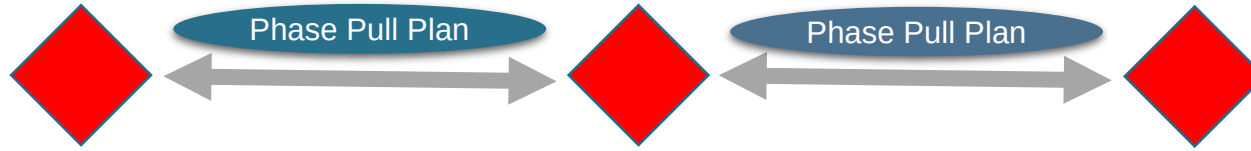


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# Milestone Planning



- Define the overall road map and gain alignment
- Identify milestones important to client and stakeholders – especially immovable dates
- Informs the Phase Pull Planning



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## Milestone Planning



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## Phase Pull Planning



### Phase Pull Planning

- Phase of the work (8-12 wks)
- Informed by the **Milestone Plan**
- Work out the structure and durations
- After – add dates and transfer to the **Look Ahead Plan**



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|                     |          |        |  |
|---------------------|----------|--------|--|
| Location            |          | Crew   |  |
| DH 1                |          | 5      |  |
| Activity            |          | I Give |  |
| UNDERSLAB SERVICES  |          |        |  |
| GL 21-25            |          |        |  |
| Predecessor / I Get |          |        |  |
| STEEL               |          | 107    |  |
| Constraint          |          |        |  |
| Tag ID              | Duration |        |  |
| 389                 | 5        |        |  |

|                     |          |        |  |
|---------------------|----------|--------|--|
| Location            |          | Crew   |  |
| DH 1                |          | 5      |  |
| Activity            |          | I Give |  |
| UNDERSLAB SERVICES  |          |        |  |
| GL 15-21            |          |        |  |
| Predecessor / I Get |          |        |  |
| STEEL               |          | 107    |  |
| Constraint          |          |        |  |
| Tag ID              | Duration |        |  |
| 390                 | 7        |        |  |

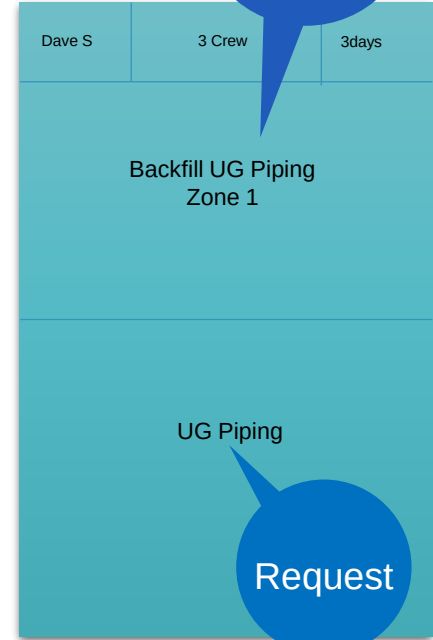
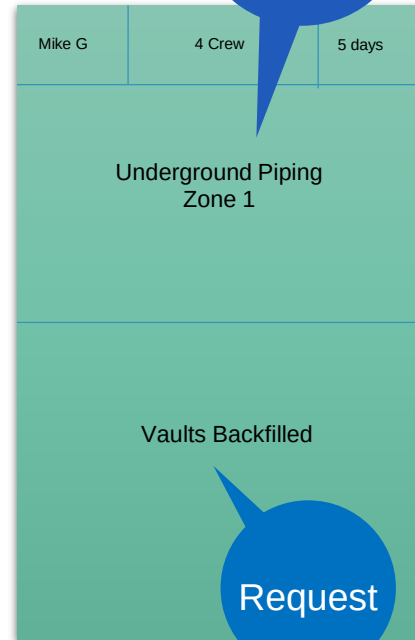
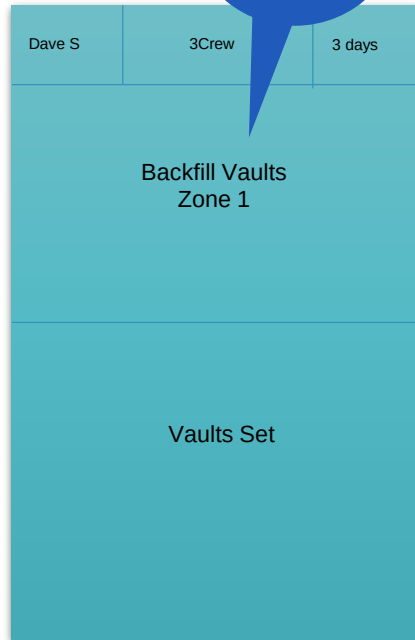


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# Pull Example



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# Start at End



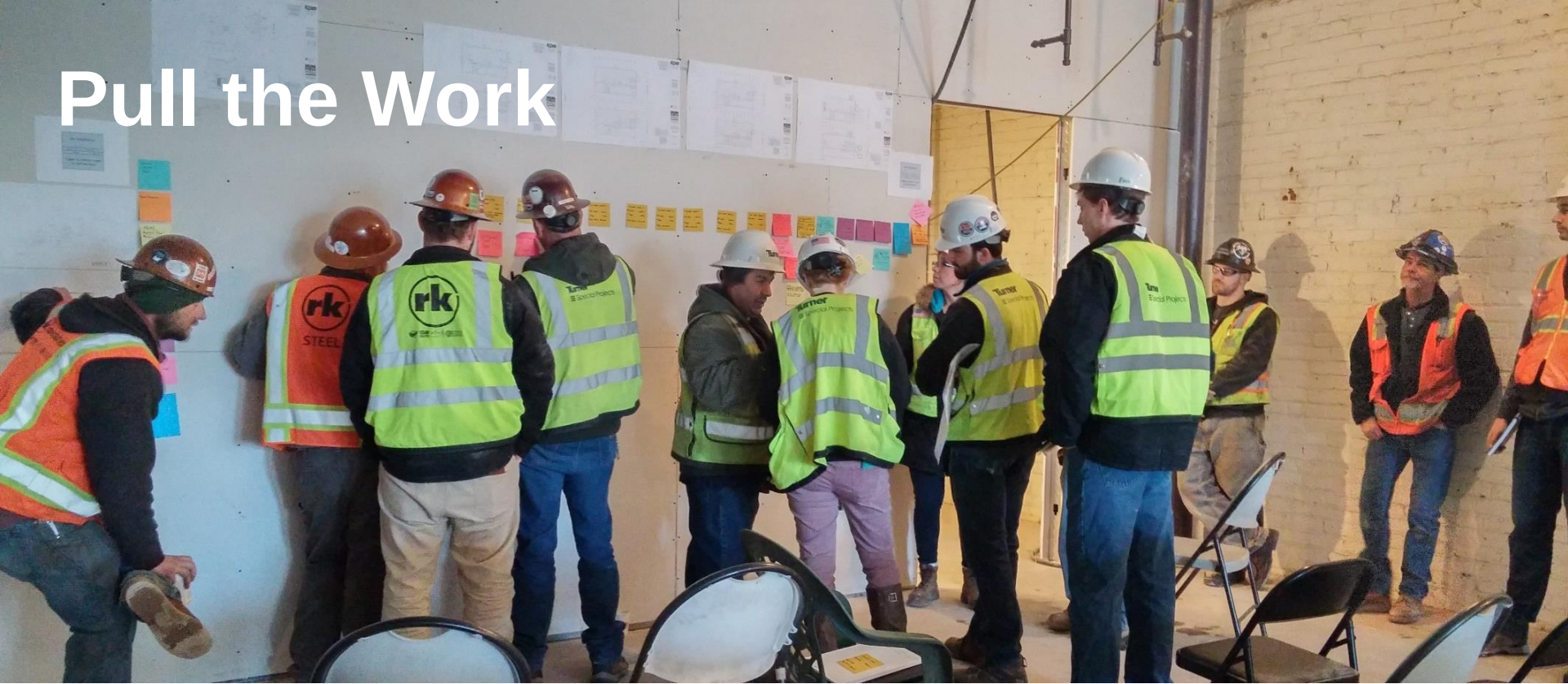
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# Pull the Work



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# Arrive at the Start



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## Look Ahead Planning

Constraint Description

Responsible Person & CO

Date Needed

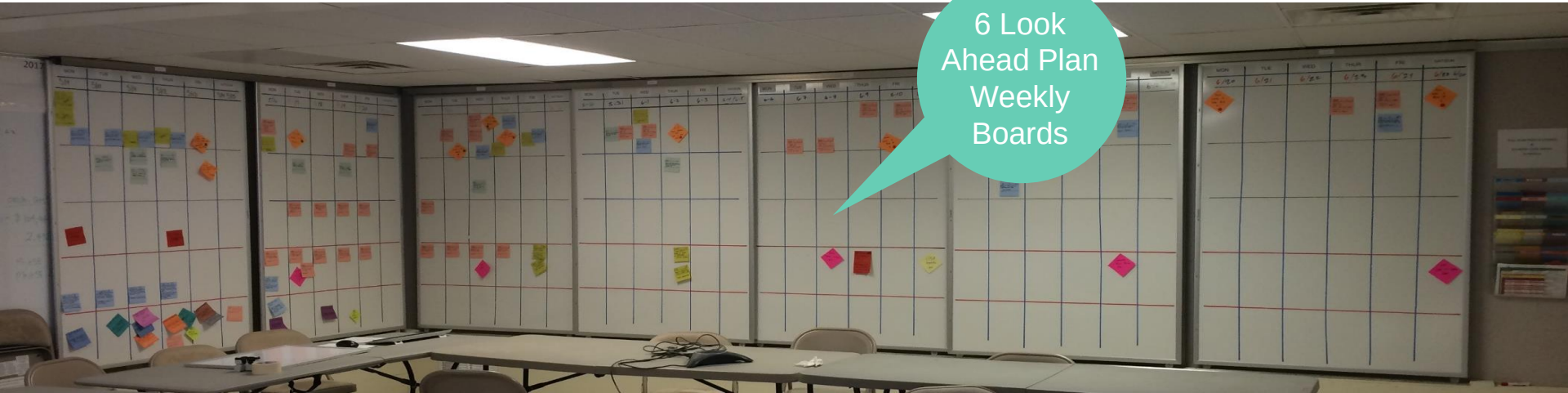
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## Look Ahead Planning

## Boards in Weekly Meeting Space



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## Weekly Work Planning



## Weekly Work Plan Informs the Daily Huddle



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## Daily Huddle

1. What did I complete?
2. What will I complete?
3. What needs to be re-planned?



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# Plan Percent Complete

Date Range: 2015-04-15 to 2015-10-13  
Report Date: 2015-10-12



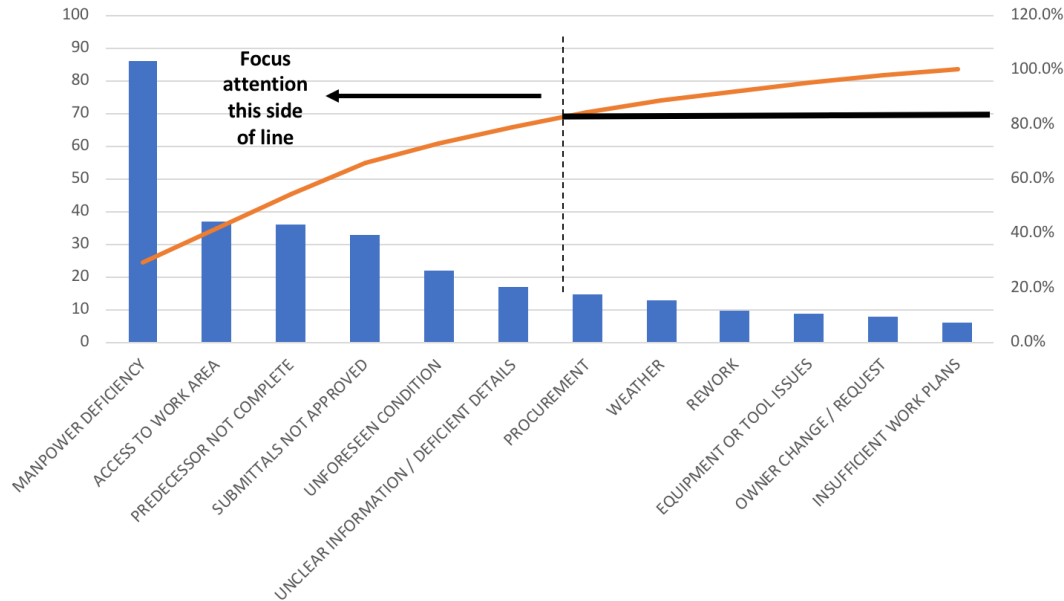
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# Reasons for Variance

Sample Variance Analysis - Missed Commitments  
Pareto Chart



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# Future Training Opportunities

- CSU Dominguez Hills - Friday, November 16, 2018
- Cal Poly San Luis Obispo – Tuesday, December 11, 2018
- CSU East Bay – Wednesday, February 27, 2019
- CSU Sacramento – Friday, March 15, 2019



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# Questions?

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